

Figure 3: Menu with multiple-selection 'checked' items

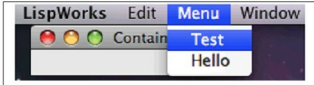


Figure 4: Individual menu

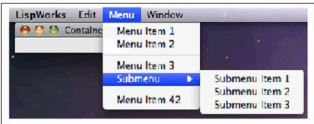


Figure 5: Menu hierarchy

keyword, selectable menu items—either as multiple-selection or single-selection items. This is like having radio buttons or check boxes as items in a menu, and is a popular technique among many GUI-based applications. The following example shows you how to include a panel of radio buttons in a menu (see Figure 2):

```
(setq radio (make-instance 'menu-component
    :interaction :single-selection
    :items '("This" "That")
    :callback 'hello))

(setq commands (make-instance 'menu
    :title "Commands"
    :items
    (list "Command 1" radio "Command 2")
    :callback 'test-callback))

(contains commands)
```

The menu items *This* and *That* are radio buttons, only one of which may be selected at a time. The other items are just ordinary commands, as in the previous example. Note that CAPI automatically groups items that are parts of a menu component, so that they are separated from other items in the menu.

Checked menu

The above example also illustrates the use of more than one callback in a menu, which of course is the usual case

when you are developing real applications. Choosing either of the radio buttons displays one message on the screen, and choosing either *Command1* or *Command2* returns the arguments of the callback.

Checked menu items can be created by specifying *:multiple-selection* to the *:interaction* keyword, as illustrated below (view Figure 3):

```
(setq letters (make-instance 'menu-component
    :interaction :multiple-selection
    :items (list "Alpha" "Beta")))

(contains (make-instance 'menu
    :title "Greek"
    :items (list letters)
    :callback 'test-callback))
```

Note how the items in the menu component inherit the callback given to the parent, eliminating the need to specify a separate callback for each item or component in the menu. Within a menu or component, you can specify alternatives that are invoked by modifier keys for a main menu item.

The *menu-item* class lets you create individual menu items, which can be passed to menu-components or menus via the *:items* keyword. Using this class, you can assign different callbacks to different menu items. Remember that each instance of a menu item must not be used in more than one place at a time.

```
(setq test (make-instance 'menu-item
    :title "Test"
    :callback 'test-callback))

(setq hello (make-instance 'menu-item
    :title "Hello"
    :callback 'hello))

(setq group (make-instance 'menu-component
    :items (list test hello)))

(contains group)
```

The combination of menu items, menu components and menus can create a hierarchical structure. The menu in the below code has five elements, one of which is itself a menu (with three menu items) and the remainder are menu components and menu items. Items in a menu inherit values from their parent, allowing similar elements to share relevant properties whenever possible.

```
(defun menu-item-name (data)
    (format nil "Menu Item -0" data))

(defun submenu-item-name (data)
```